

3D Coupled Modelling of a Series-Connected Bifilar Cu-Stabilised REBCO Resistive SFCL

H-formulation, solid heat transfer and external circuit solved together on a 50 mm representative segment of four tape runs, 400 V source model

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84.2 %

independent peak current reduction
(the two peaks are 3.6 ms apart)

79.37 K

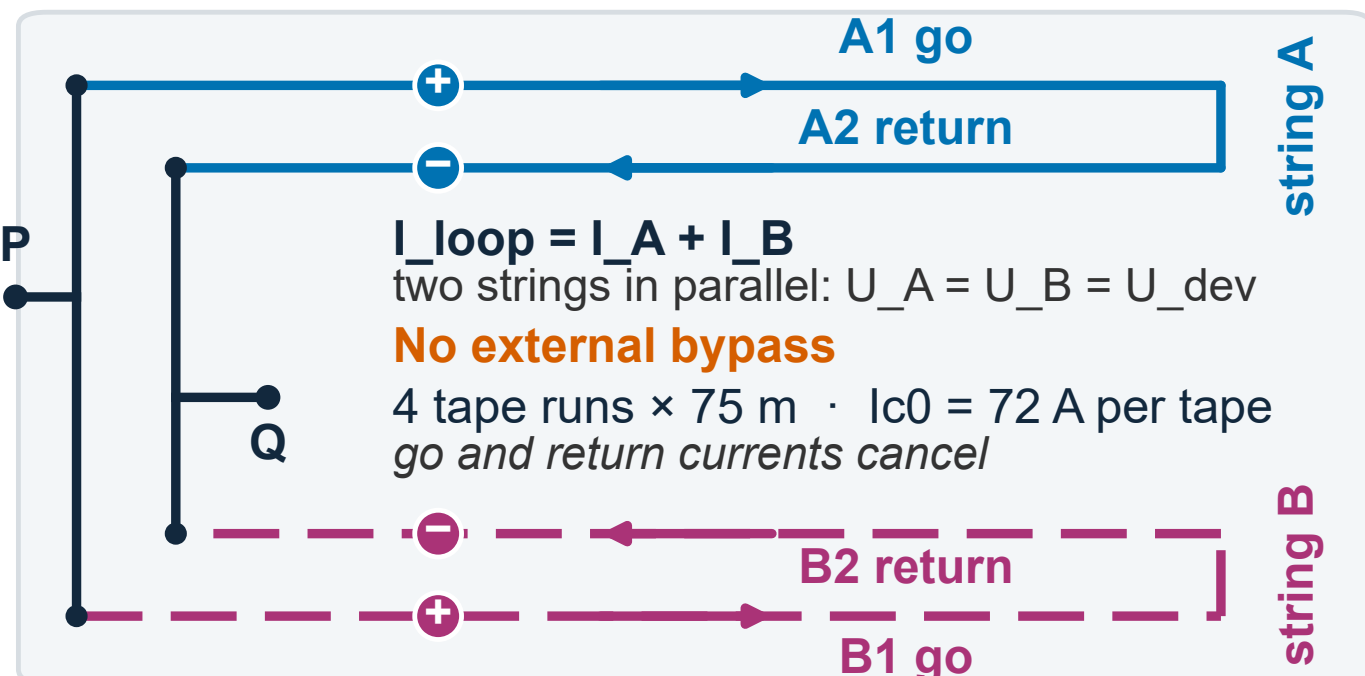
volume mean peak in the 20 ms window;
local extremum 79.67 K, mesh sensitive

309.3 V

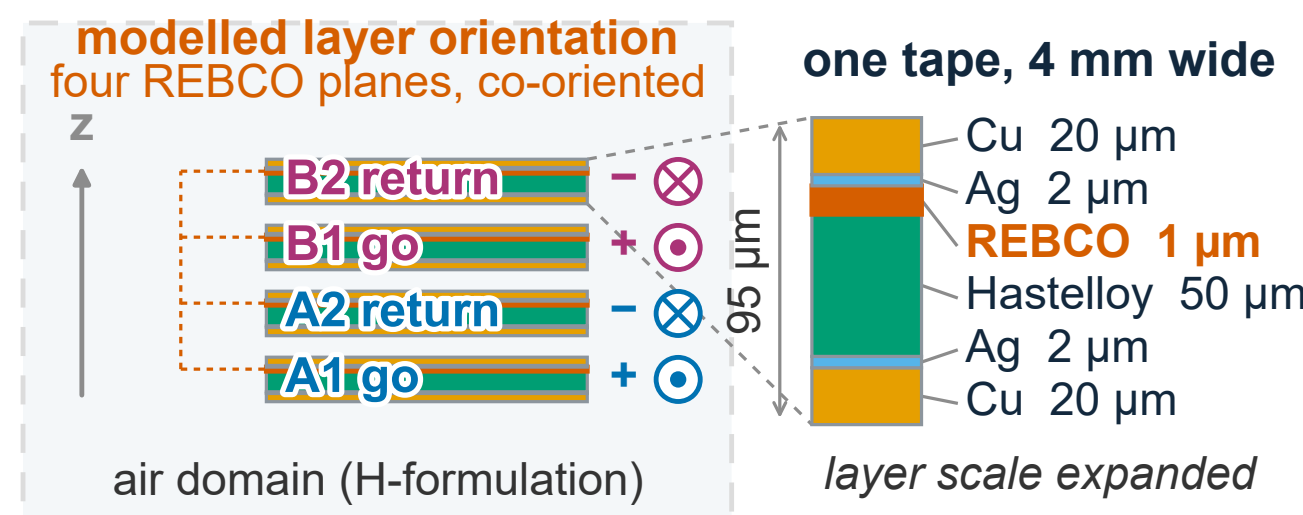
voltage peak across the device,
0.4 ms after the current peak

1 Device topology and length scaling

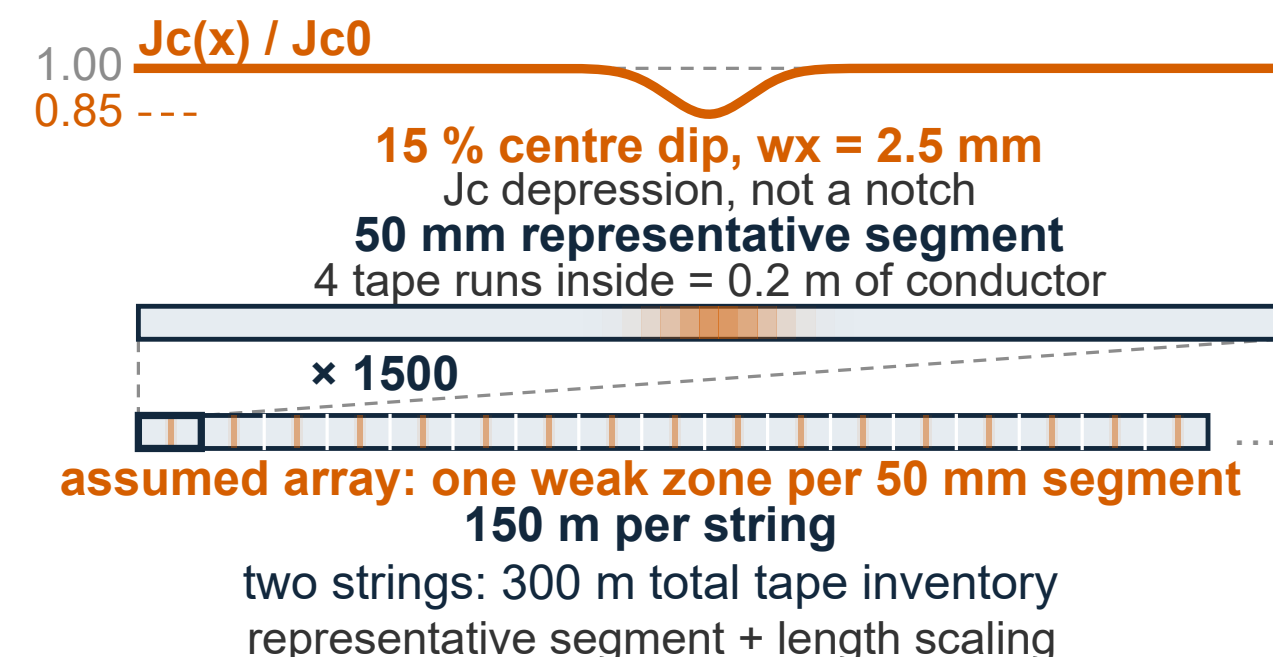
(a) BCS electrical topology



(b) Stacked go-return pairs and conductor layer order

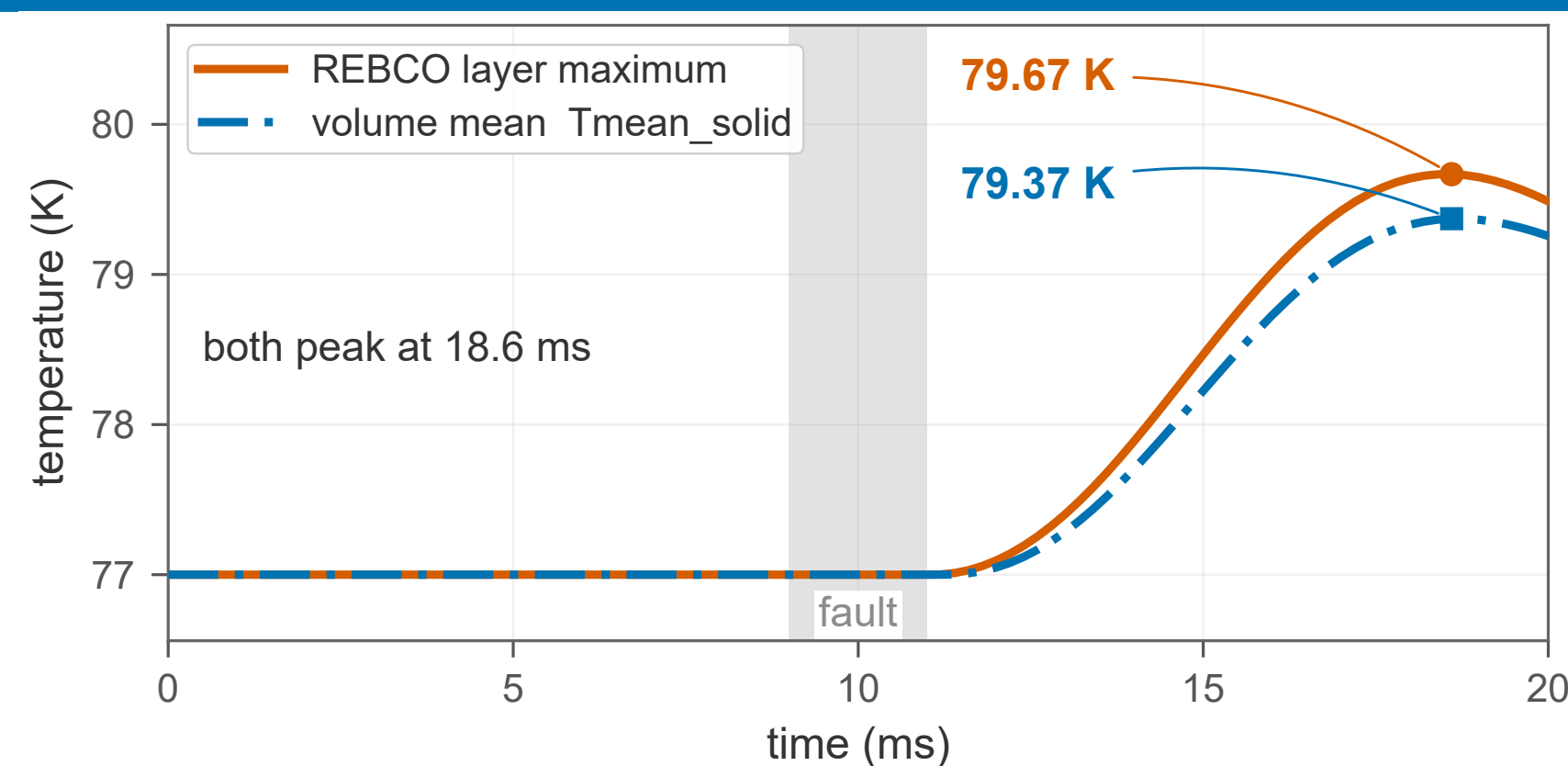
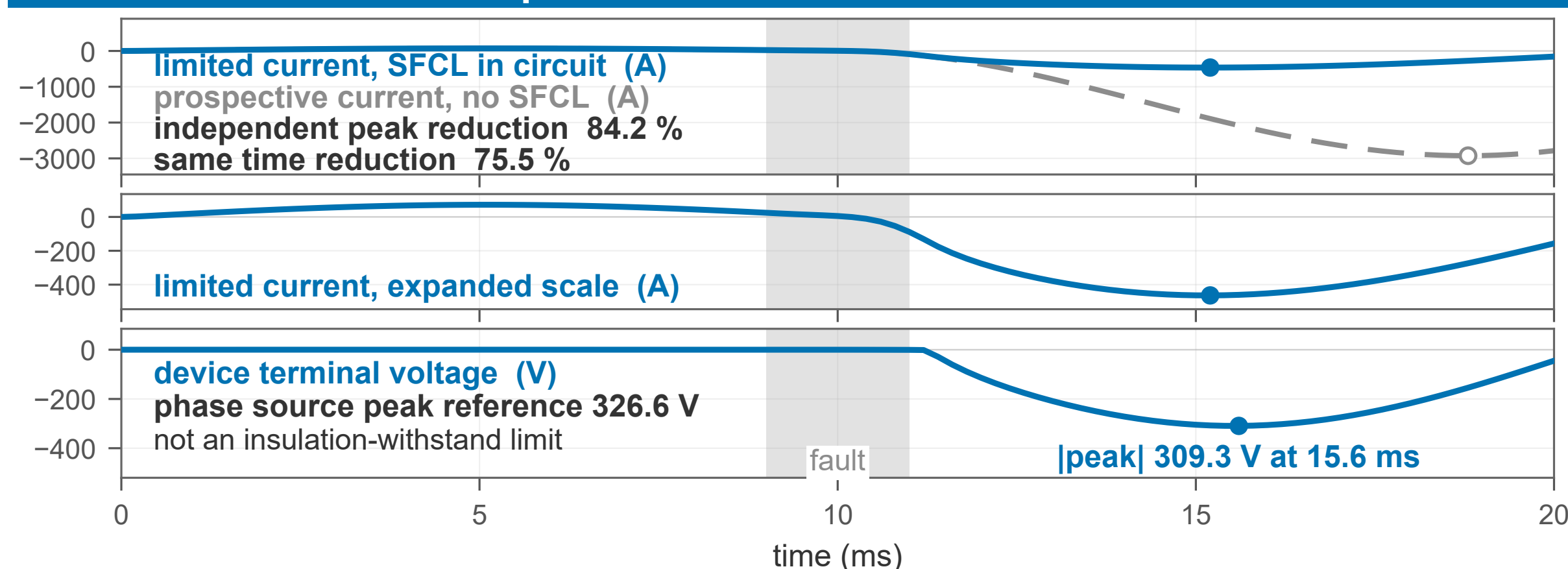


(c) Segment → assumed array → device



Two hairpin strings in parallel; in each, outgoing and return tapes are in series so their currents cancel. Panel (b) is the four stacked tapes, panel (c) the modelled segment and its length scaling.

2 Circuit and thermal response

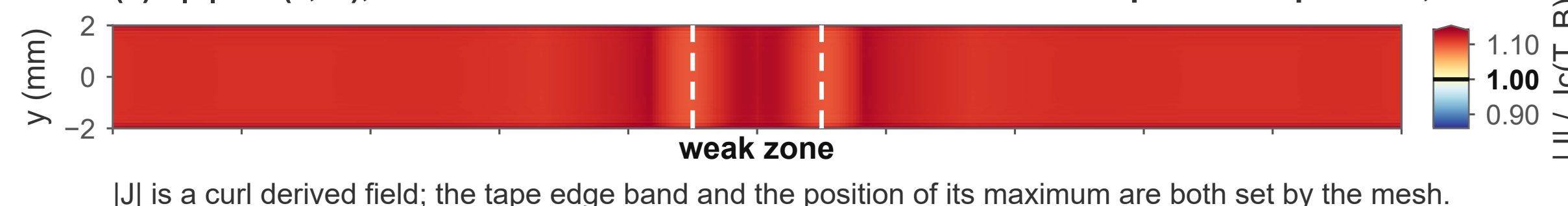


Current limiting and device terminal voltage on one shared time axis. The voltage peak lags the current peak, so the two are not one operating point; the modelled peaks are negative and the sign carries no design meaning.

Temperature, double reported: the volume mean over the 24 solid domains, and the extremum, a maximum over volume value on the $1 \mu\text{m}$ REBCO layer that is never a design point value on its own.

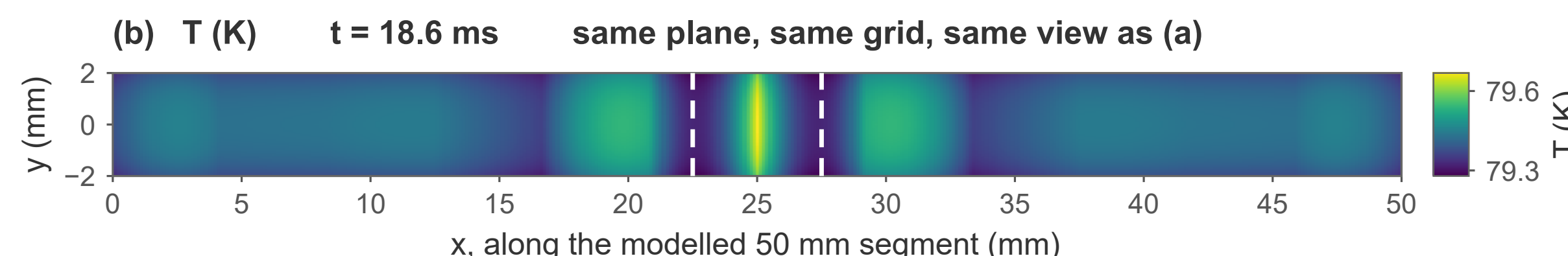
3 Spatial electromagnetic and thermal fields

(a) $|J| / J_c(T, B)$, dimensionless $t = 15.2 \text{ ms}$ REBCO mid-plane of tape run 4, $z = 0.6575 \text{ mm}$



the whole layer is above unity at this instant: 100.00 % of the sampled plane
weak zone: centre $x = 25 \text{ mm}$, $w_x = 2.5 \text{ mm}$, minimum $J_c/J_{c0} = 0.85$
(a) plane maximum is a sampled lower bound on the extremum over the domain

plane max 1.2309
on a tape edge row, $|y| = 2.00 \text{ mm}$; x position set by the mesh



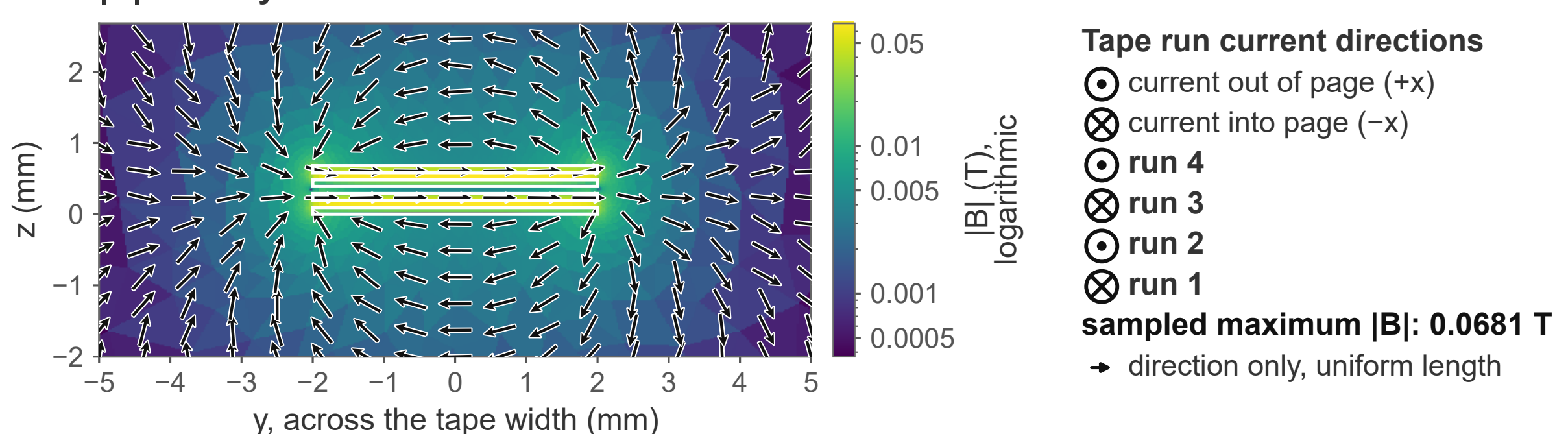
plane max (grid-sampled) 79.66796806 K · mean 79.41655240 K
min 79.27894020 K

Reference-mesh field; hotspot magnitude remains mesh-sensitive.

plane maximum at $x = 25.00 \text{ mm}$, $y = 0.00 \text{ mm}$

The same plane, same grid, at the two instants the figure names: normalised current density above, temperature below. Dashed lines mark the weak zone, and both panels share one axial axis.

$|B|$ on the y-z section at $x = 25.0 \text{ mm}$ $t = 15.2 \text{ ms}$



Tape run current directions

- ⊙ current out of page (+x)
 - ⊗ current into page (-x)
 - ⊙ run 4
 - ⊗ run 3
 - ⊙ run 2
 - ⊗ run 1
- sampled maximum $|B|$: 0.0681 T
→ direction only, uniform length

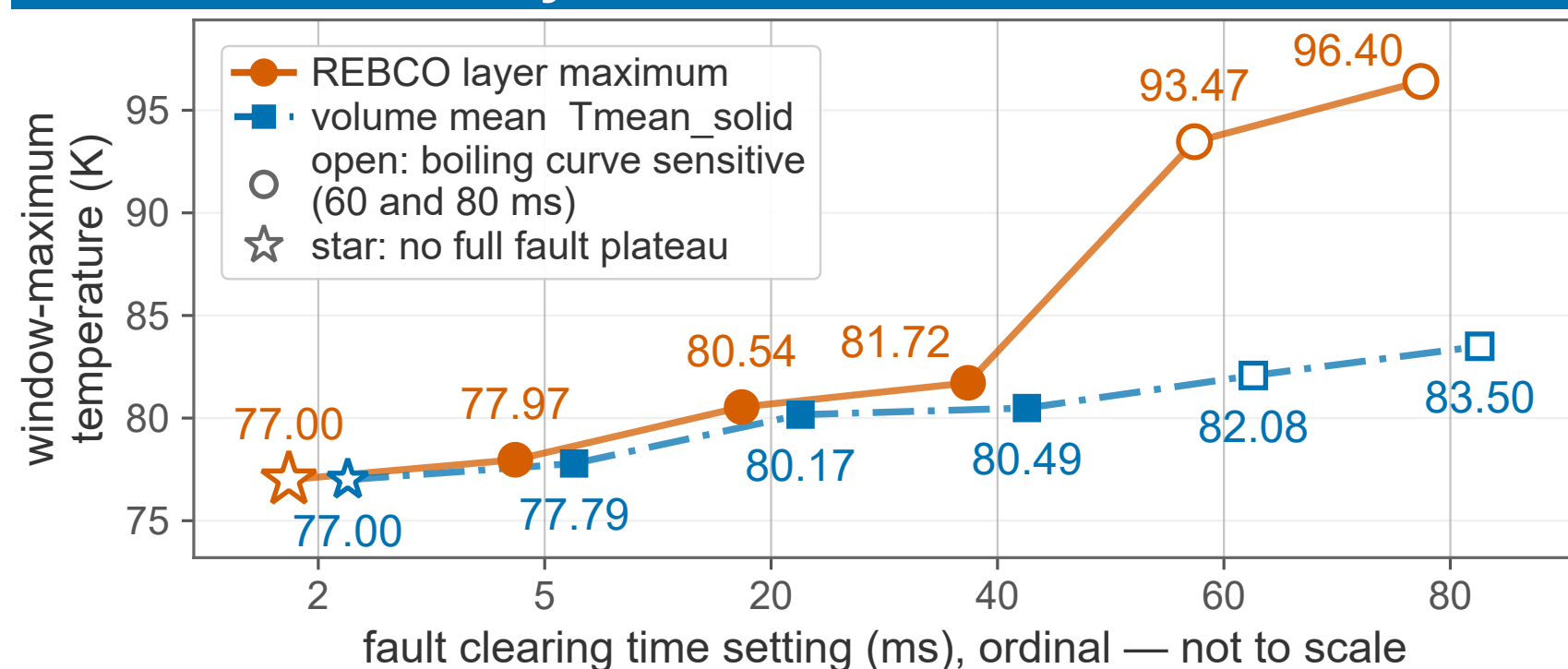
share of window E·J energy (%)



Share of the modelled solid $E \cdot J$ energy over the window, by material group. It is a segment-level time integral, not an instantaneous power split, and length scaling does not change it.

Field magnitude on the transverse section through the weak zone, with in-plane direction arrows.

4 Model credibility and limitations

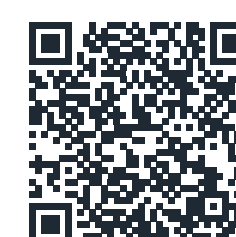


Six discrete breaker settings, one run each. The calibres differ between settings, so the six are not one homogeneous scan, nothing between them is interpolated, and the two longest remain unresolved.

Take home. Within this model line the device holds the line current far below the prospective fault current, the solid stays within about three kelvin of the bath, and most of the window $E \cdot J$ energy goes to the copper stabiliser.

References

- [1] W. Song et al., IEEE Trans. Transp. Electr. 7 (2021) 276–286. doi:10.1109/TTE.2020.2998417
 - [2] B. Shen et al., Supercond. Sci. Technol. 33 (2020) 033002. doi:10.1088/1361-6668/ab66e8
 - [3] N. Riva et al., Supercond. Sci. Technol. 33 (2020) 114008. doi:10.1088/1361-6668/aba34e
 - [4] B. V. Balakin et al., Cryogenics 65 (2015) 5–9. doi:10.1016/j.cryogenics.2014.11.003
- Solver settings, repeats and core counts: see the appendix.



TODO: replace QR_TARGET